

Tezpur University

Semester End Examination, Spring 2025

CS417 : OPERATING SYSTEM

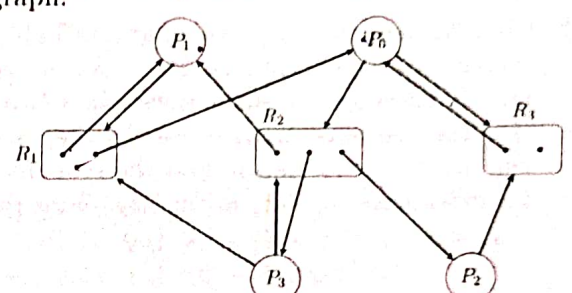
Max. Marks: 50

Max. Time: 2 hrs.

All questions are compulsory except otherwise stated.

Figures in the right margin indicate mark(s) allotted to the individual/groups of question(s).

Questions 1 through 10 are MCQ type questions. Choose the best option and re-write the option number as well as the answer. $10 \times 2 = 20$

1. On a system that implement multilevel feedback queues, a totally CPU-bound process requires 40 time-units to execute. If the first queue uses a time quantum of 2 time-units and at each level the time quantum is increased by 5 time-units, how many times will the job be interrupted?
 (A) 2 (C) 4
 (B) 3 (D) 5
2. On a simple paging system with 2^{24} bytes of physical memory, 256 pages of logical address space, and a page size of 2^{10} bytes. How many bits are there in the logical address?
 (A) 10 (C) 16
 (B) 12 (D) 18
3. On a system using a disk cache, the mean access time is dependent on the mean cache access time, the mean disk access time, and the hit rate. Calculate the mean access time, given: cache access time = 1 ms, disk access time = 100 ms, hit rate = 82%.
 (A) 82.18 ms (C) 18.82 ms
 (B) 54.81 ms (D) 26.28 ms
4. At a particular time of computation, the value of a counting semaphore is 7. Then, 20 P (proberen) operations and 'x' V (verhogen) operations were completed on this semaphore. If the final value of the semaphore is 5, 'x' will be
 (A) 15 (C) 22
 (B) 18 (D) 25
5. A process during a certain time in its execution refers to the following 5 pages—A, B, C, D, and E in the following order: A, B, C, D, A, B, E, A, B, C, D, and E. How many page faults will it generate if 3 frames are given to each processes in an internal page replacement scheme?
 (A) 7 (C) 9
 (B) 8 (D) 10
6. If the mean latency in disk access is only measured in average rotational latency and average seek time. What is the mean latency of a popular hard disk with rating of 7200 RPM and 8.5 ms of disk rotation speed and average track seek time, respectively? Average rotational latency may be computed as half the time taken for a full revolution.
 (A) 9.60 ms (C) 11.56 ms
 (B) 10.55 ms (D) 12.66 ms
7. Consider the following resource allocation graph:

 (A) The system is in safe state
 (B) The system is deadlocked
 (C) The system is in unsafe state
 (D) The system is in an unpredictable state
8. Assuming that every entry in the page table of a system with 32-bit long logical address space and 4 KiB page size is 4 bytes, what will be the amount of physical memory required to store the page table of such a system?
 (A) 2 MiB (C) 8 MiB
 (B) 4 MiB (D) 16 MiB
9. Consider the following snippet of C code written to take advantage of parallel processing using OpenMP. How does OpenMP run/execute the code in a quad core CPU?

```
#pragma omp parallel for
for (i = 0; i < N; i++) {
    c[i] = a[i] + b[i];
}
```

- (A) All 4 cores individually operate on the entire data for redundancy and data safety
 - (B) The N data is equally split to all the 4 cores for speed
 - (C) The data is split into half and a pair of core work on the entire half for speed and redundancy
 - (D) Only 1 core will operate on the data as delegation primitives are not provided
10. A system has 3 processes sharing 4 resources. If each process needs a maximum of 2 units of resources then
- (A) deadlock can never occur
 - (B) deadlock may occur
 - (C) deadlock has to occur
 - (D) cannot decide

For questions 11 through 16, answer as directed.

11. How does the concept of implicit threading help developers produce better software? Provide a proper justification. How are Java threads different from C/C++ OpenMP library? $1 + 2 + 2 = 5$

OR

Explain the memory management mechanism adopted by the Intel 32-bit IA-32 architecture. Will it be sufficient in case of Intel 64-bit x86-64 architecture. Justify your answer briefly. $3 + 2 = 5$

12. Give examples of operating systems built based on monolithic structure and microkernel approaches. Comment on program/module integration and on extensibility of the operating system of the two approaches. List some examples of difficulties that disadvantaged the layered approach. $1 + 2 + 2 = 5$
13. Consider `int A[100][100] = new int[100][100];` and `A[0][0]` is at location 200 in a paged memory system with pages of size 200. Determine how many page faults are generated by the following matrix initialization loops, if we follow LRU on a 3 frame internal page replacement scheme. Assume that the first page frame is used by the logic of the process, Page 0 (location 0 to 199), that manipulate the matrix initialization, and the remaining two frames are empty initially. Discuss the reason(s) for differences (or not) in the page fault rate. $3 + 2 = 5$

(a) `for (int j = 0; j < 100; j++)
 for (int i = 0; i < 100; i++)
 A[i][j] = 0;`

(b) `for (int j = 0; j < 100; j++)
 for (int i = 0; i < 100; i++)
 A[i][j] = 0;`

OR

A system that uses a two-level page table has pages of size 4 KiB and a 32-bit virtual address space. The first 8-bit of the logical address serve as the index to the first-level page table. Find the number of - $5 \times 1 = 5$

- (a) bits that specify the page offset.
 - (b) bits that specify the second-level index.
 - (c) entries in the level-one page table.
 - (d) entries in the level-two page table.
 - (e) pages in the virtual address space.
14. Given that you have a choice to build an operating system either using deadlock prevention mechanism to address deadlock conditions or to simply allow deadlock to occur and activate recovery mechanisms. Which option will you choose and why? How is deadlock prevention different from the deadlock avoidance mechanism? $3 + 2 = 5$

OR

Discuss how is segmentation different from paging. With an example show how a user program that has a function and the main driver function (assuming C language) is allocated memory and hence the segment table is formed. Assume that the primary memory has enough holes scattered to fit the entire program's logical address space but not in contiguous form. $2 + 3 = 5$

15. For some systems the page tables are too large to have a hardware. In such systems the page tables are, often, maintained in the main memory. How does a TLB (translation look-aside buffer) help reduce the frequency of accessing page table and the pages referred by the CPU from the memory? Justify your answer with a proper example. Comment on wired down TLB entries. $3 + 2 = 5$
16. Assuming that 70 percent of the components of a process are parallelly computable. What is the gain in terms of speed up that we may get if we go from 1 core to 4 cores. What will be the scenario if we can significantly increase the number of cores, say infinite? $3 + 2 = 5$